Polynomial-time simulation of non-Clifford quantum error correction

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arXiv

2610.06811

Overview

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1 **Most magic-state preparation protocols**

Circuits made of CNOT, T, Paulis, Z measurements and *compatible* X measurements.

Examples:

- Magic state distillation and cultivation based on applying a logical 3rd level non-Clifford gate to a code.
- Code switching between 3D and 2D codes.

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

1. Most magic-state preparation protocols
2. **Simulated**

Maintain state representation throughout the circuit.
Two representations: Schrödinger (phase polynomials) and Heisenberg (generalized stabilizer group).

Can compute the following:

- Pauli expectation values and amplitudes
- Measurement samples
- State equality

Overview

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1. Most magic-state preparation protocols
2. Simulated
3. **Polynomial time**

Polynomial-time complexity in the number of qubits and circuit depth.

No exponential dependence in the number of T gates!

No exponential in the number of logical qubits, in contrast to previous work[1].

[1] S. Surti, L. Daguerre, and I. H. Kim. *Efficient simulation of logical magic state preparation protocols* (2026).

Outline

1. The phase polynomial representation
2. Compatible measurements
3. DCP stabilizer formalism
4. Benchmarking our simulator

1 The phase polynomial representation

Phase-polynomial states

Unitary gate set: the third-order CNOT-dihedral group $G_3 = \langle \text{CNOT}, T, X \rangle$[2]

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Question: what happens when we apply G_3 gates to $|0\rangle^{\otimes \ell} |+\rangle^{\otimes m}$?

$$U |0\rangle^{\otimes \ell} |+\rangle^{\otimes m} \propto \sum_{y \in \mathbb{F}_2^m} U |x\rangle$$

with $|x\rangle = |0\rangle^{\otimes \ell} |y\rangle$

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New question: what is $U |x\rangle$ for $U \in G_3$?

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$$T_i\,|x\rangle = \omega^{x_i}\,|x\rangle$$

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for a so-called **third-order phase polynomial**:

$$f(x) = \sum_i a_i x_i + 2\sum_{i<j} b_{ij} x_i x_j + 4\sum_{i<j<k} c_{ijk} x_i x_j x_k$$

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with S an affine subspace of $\mathbb{F}_2^{n+m}$ and q a third-order phase polynomial.

Phase-polynomial states

Definition (3PP state): a third-order phase polynomial (3PP) state is a state of the form

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Question: but what about measurements?

2 Compatible measurements

Compatible measurements: Z measurements

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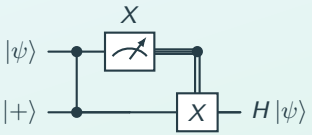
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$\Rightarrow$ Still a 3PP state!

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Observation 1: not every X measurement can be compatible.

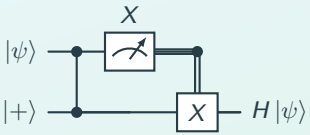
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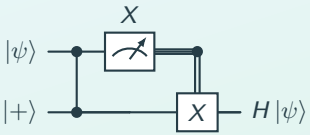


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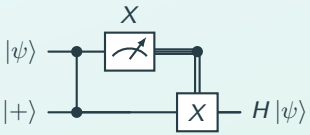
$$H|0\rangle = |+\rangle \quad \text{(compatible)}, \qquad H|T\rangle \propto \cos\tfrac{\pi}{8}|0\rangle - i\sin\tfrac{\pi}{8}|1\rangle \quad \text{(incompatible)}.$$

The latter has unequal amplitude magnitudes, so it is not a 3PP state.

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Question: which X measurements are compatible with a state $|\psi\rangle$ specified by $(S, q(x))$?

Compatible X measurements: support and phase

Claim: compatibility of measuring $X(v)$ is determined by whether $S = S + v$ and, if so, by the form of $\Delta_v q(x) = q(x+v) - q(x) \pmod 8$ on S.

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If $S = S + v$: relabeling the second sum gives

$$\begin{aligned} \Pi_\mu |\psi\rangle &\propto \sum_{x \in S} \left(\omega^{q(x)} + (-1)^\mu \omega^{q(x+v)}\right) |x\rangle \\ &\propto \sum_{x \in S} \omega^{q(x)} \left(1 + \omega^{\Delta_v q(x) + 4\mu}\right) |x\rangle \end{aligned}$$

Classification of compatible X measurements

Theorem: measuring $X(v)$ on a 3PP state (S, q) is compatible iff

1. $S \neq S + v$, or
2. $S = S + v$ and one of the following holds on S (modulo 8):

(2a) $\Delta_v q(x) = 4f(x)$, f affine,

(2b) $\Delta_v q(x) = 2 + 4g(x)$, $2g$ a third-order phase polynomial,

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Proof: $X(v)$ shifts the support, so $S + v = S$. Matching amplitudes at $x \in S$ gives

$$\omega^{q(x+v)} = (-1)^{\mu}\omega^{q(x)} \;\Rightarrow\; q(x+v) = q(x) + 4\mu \pmod 8 \;\Rightarrow\; \Delta_v q(x) = 4\mu \pmod 8.$$

Thus $\Delta_v q = 4f$ with $f(x) = \mu$ constant, hence affine: case (2a).

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Proof: for $x \in S$, $Z(w)\,|\psi\rangle = |\psi\rangle$ gives $w \cdot x = 0$. Anticommutation gives $w \cdot v = 1$. Thus:

$$w \cdot (x + v) = 0 + 1 \neq 0 \pmod 2 \;\Rightarrow\; x + v \notin S \;\Rightarrow\; S \neq S + v.$$

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Noise model: DCP errors at every circuit location

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Theorem (measurement compatibility under noise): assume

1. Circuit made of CNOT-dihedral gates and compatible measurements
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Then, all measurements remain compatible under arbitrary inserted DCP faults.

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1. Circuit made of CNOT-dihedral gates and compatible measurements
2. Every ideal X measurement is either deterministic, or anticommutes with a current Z stabilizer

Then, all measurements remain compatible under arbitrary inserted DCP faults.

Examples:

- **Distillation and non-Clifford computation**: logical CNOT-dihedral gates on CSS codes, followed by ideally deterministic checks.
- **DCP checks in cultivation**: controlled-DCP gates give deterministic ancilla X readouts.
- **CSS gauge fixing and code switching**: new X checks are fixed or anticommute with a current Z constraint.

3 DCP stabilizer formalism

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Observation: if we start with a state stabilized by a Pauli operator S, and apply a gate $U \in G_3$, then the new state is stabilized by $S' = USU^\dagger$, **which is a DCP operator**.

$\Rightarrow$ Let's look at states uniquely stabilized by a group of DCP operators. We call these **DCP states**.

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Example: $|00\rangle$ is stabilized by $\mathcal{S}_{\text{DCP}} = \langle S_1, S_2, \text{CZ}_{12} \rangle$. Interestingly:

1. It has $m = 3$ independent generators, but $n = 2$ qubits. Contrasts with Pauli case, where $n = m$.
2. However, also uniquely stabilized by some dim-2 subgroups of $\mathcal{S}_{\text{DCP}}$: $\langle Z_1, Z_2 \rangle$, $\langle Z_1, S_2 \rangle$, etc.

Questions:

1. What is the link between DCP states and 3PP states?
2. Is every DCP state uniquely stabilized by an n-dimensional DCP subgroup?
3. Can we build a *DCP tableau simulator*, with update rules for compatible measurements?

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1. Every 3PP state can be written as $|\psi\rangle = U\left(|0\rangle^{\otimes \ell} |+\rangle^{\otimes m}\right)$ for some $U \in G_3$.
2. $|0\rangle^{\otimes \ell} |+\rangle^{\otimes m}$ is uniquely stabilized by $\langle Z_1, \ldots, Z_\ell, X_{\ell+1}, \ldots, X_{\ell+m}\rangle$.
3. $|\psi\rangle$ is stabilized by the DCP operators $\langle UZ_1U^\dagger, \ldots, UZ_\ell U^\dagger, UX_{\ell+1}U^\dagger, \ldots, UX_{\ell+m}U^\dagger\rangle$, and one can prove it's unique.

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DCP states are 3PP states: let G the DCP stabilizer group of $|\psi\rangle$. We note its X-part as:

$$X(G) = \{v : DX(v) \in G \text{ for some diagonal } D\}.$$

Support: $S = x_0 + X(G)$, for a vector x_0 fixed by the diagonal stabilizers.

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Phases: the diagonal factor in $DX(v)$ fixes the relative phase between x and $x + v$.

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Therefore: compatibility tests and tableau updates take polynomial time.

4 Benchmarking our simulator

MERLIN: from theory to benchmarks

MERLIN: our open-source simulator for non-Clifford QEC

Input: Stim-like circuit descriptions.

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Benchmark circuits:

- Bravyi–Haah distillation: $3k + 8 \to k$ magic states, increasing k.
- 15-to-1 distillation and distance-3 / distance-5 cultivation.
- CCZ code switching with $[\![27, 3, 3]\!]$ and $[\![81, 3, 5]\!]$ codes[3].

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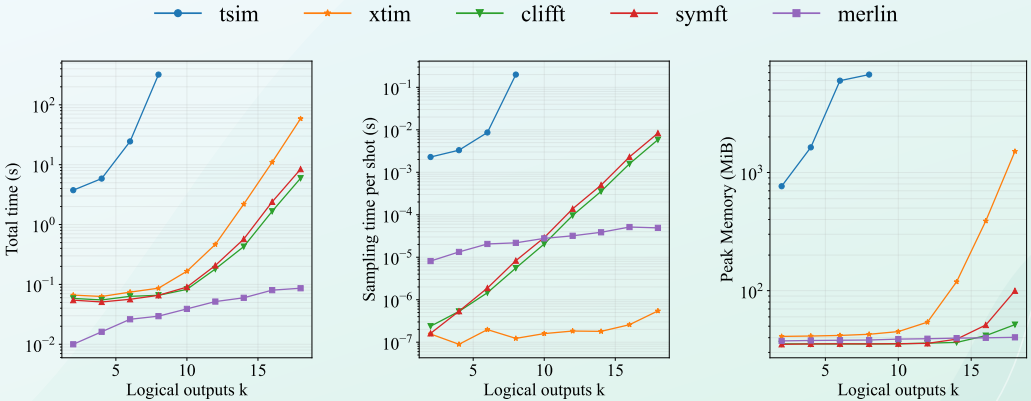
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Metrics: total runtime, sampling time per shot and peak memory.

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Bravyi–Haah: scaling with logical outputs



1000 attempted shots, $p = 10^{-3}$.

Total includes pre-processing. Sampling time excludes pre-processing.

Distillation, cultivation and code switching

Total runtime in ms, including pre-processing

Circuit	Tsim	Xtim	Clifft	SymFT	**MERLIN**
15-to-1	5029.95	85.86	99.56	57.67	**12.52**
Cultiv-3	OOM	65.29	54.29	63.63	**28.98**
Cultiv-5	OOM	253.51	**67.93**	577.20	202.01
Switch-3	OOM	2817.53	OOM	ERR	**573.69**
Switch-5	OOM	OOM	OOM	ERR	**6959.13**

Time per shot in μs, excluding pre-processing

Circuit	Tsim	Xtim	Clifft	SymFT	**MERLIN**
15-to-1	3446.85	0.40	0.37	**0.18**	9.62
Cultiv-3	OOM	0.39	0.51	**0.34**	28.34
Cultiv-5	OOM	7.70	7.97	**6.52**	200.15
Switch-3	OOM	**2.65**	OOM	ERR	549.93
Switch-5	OOM	OOM	OOM	ERR	**6704.94**

1000 attempted shots, $p = 10^{-3}$. Cultivation excludes the escape stage.
OOM = allocation above 20 GB. ERR = runtime error. Cultiv / Switch suffix = distance.

Discussion: summary

In summary:

1. We have shown that CNOT-dihedral gates generate a class of states, which can be characterized by polynomial phases in the Schödinger picture, and DCP stabilizers in the Heisenberg picture.

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2. We have classified all the measurements that are compatible with this class, making it useful for QEC.
3. We have implemented a simulator, MERLIN, that can simulate these circuits in polynomial time, and benchmarked it against other simulators.

Discussion: parallel work

Lots of parallel work on the same topic, with different emphases and approaches:

1. **Takada, Bartlett and Williamson**[8]: Clifford-stabilizer simulation of noisy magic states, closely related to our tableau approach. **Our work**: we also give a phase-polynomial description.

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3. **Fechisin et al.**[10]: 3PP and DCP equivalence, generalized to Clifford-hierarchy. **Our work**: we focus on circuit simulation.
4. **Xiao et al.**[11]: polynomial-time syndrome and fidelity estimation for high-rate distillation, avoiding logical measurements. **Our work**: more general X measurements.

Our main distinguishing contribution: full classification of X measurements preserving 3PP states.

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Discussion: future work

1. There are still a few circuits that evade our simulation method (e.g. H-based cultivation, some more complicated distillation protocols, etc.). Can we extend our method to cover these cases?
2. Can we formalize a notion of **"DCP detector error model"** for non-Clifford circuits?
3. Can we build a theory of **"DCP codes"**, generalizing XP codes?
4. Can we construct an interesting **resource theory** based on 3PP states? A "DCP polytope" generalizing the stabilizer polytope?
5. Is there any application to **learning theory**?
6. What concrete non-Clifford protocols should we start comparing and benchmarking using Merlin?

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